Sample Paper 10

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

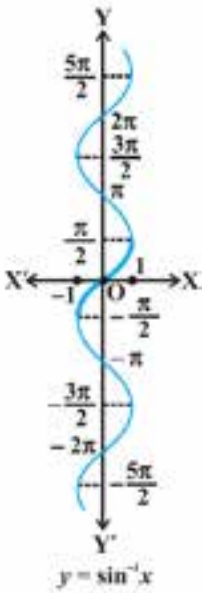
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the domain of the function $y = \cos^{-1}x$ as shown in the graph?



(a) $(-\infty,\infty)$ (b) $[-1,1]$

(c) $[0,\pi]$ (d) $[-\pi,\pi]$

2. Solution of $(x+2y^3)\,dy = y\,dx$ is

(a) $x = y^3 + cy$ (b) $x + y^3 = cy$

(c) $y^2 - x = cy$ (d) none of these

3. If $\vec{a}$ is perpendicular to $\vec{b}$ and $\vec{p}$ is a non-zero vector such that $p\vec{r} + (\vec{r}\cdot\vec{b})\vec{a} = \vec{c}$, then $\vec{r}$ is equal to

(a) $\frac{\vec{c}}{p} - \frac{(\vec{b}\cdot\vec{c})\vec{a}}{p^2}$ (b) $\frac{\vec{a}}{p} - \frac{(\vec{c}\cdot\vec{a})\vec{b}}{p^2}$

(c) $\frac{\vec{b}}{p} - \frac{(\vec{a}\cdot\vec{b})\vec{c}}{p^2}$ (d) $\frac{\vec{c}}{p^2} - \frac{(\vec{b}\cdot\vec{c})\vec{a}}{p}$

4. The lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$

and $\frac{x}{2} = \frac{y+2}{2} = \frac{z-3}{-2}$ are

(a) Parallel (b) Intersecting

(c) At right angle (d) None of these

5. A coin and six faced die, both unbiased, are thrown simultaneously. The probability of getting a head on the coin and an odd number on the die is

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$

(c) $\frac{1}{4}$ (d) $\frac{2}{3}$

6. The number of solutions of $y' = \frac{y+1}{x-1}, y(1) = 2$ is

(a) Zero (b) One

(c) Two (d) Infinite

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7. Solution set of the inequality $x \geq 0$ is

(a) half plane on the left of y-axis

(b) half plane on the right of y-axis excluding the points of y-axis

(c) half plane on the right of y-axis including the points on y-axis

(d) none of the above

8. Find the area of a curve $xy = 4$, bounded by the lines $x = 1$ and $x = 3$ and X-axis.

(a) $\log 12$ (b) $\log 64$

(c) $\log 81$ (d) $\log 27$

9. The derivative of $\cos^3 x$ with respect to $\sin^3 x$ is

(a) $-\cot x$ (b) $\cot x$

(c) $\tan x$ (d) $-\tan x$

10. Let $f(x) = x - \cos x, x \in R$, then f is

(a) A decreasing function

(b) An odd function

(c) An increasing function

(d) None of the above

11. The length of the longest interval, in which $f(x) = 3\sin x - 4\sin^3 x$ is increasing is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$

(c) $\frac{3\pi}{2}$ (d) π

12. $\int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x} dx$ is equal to

(a) 0 (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) π

13. If $f(\theta) = \begin{vmatrix} \cos^2\theta & \cos\theta\sin\theta & -\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta & \cos\theta \\ \sin\theta & -\cos\theta & 0 \end{vmatrix}$, then for all values of θ

(a) $f(\theta) = 0$ (b) $f(\theta) = 1$

(c) $f(\theta) = -1$ (d) None of these

14. If $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$ and A^2 is the identity matrix, then x is equal to

(a) -1 (b) 0

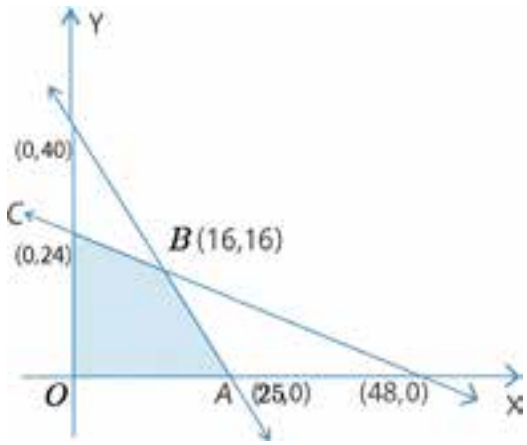
(c) 1 (d) 2

15. If $f : [0, \frac{\pi}{2}] \to [0, \infty]$ be a function defined by $y = \sin(\frac{\pi}{2})$, then f is

(a) injective (b) subjective

(c) bijective (d) none of these

16. For the feasible region in the figure, determine the maximum value of the objective function $z = 3x + 5y$.



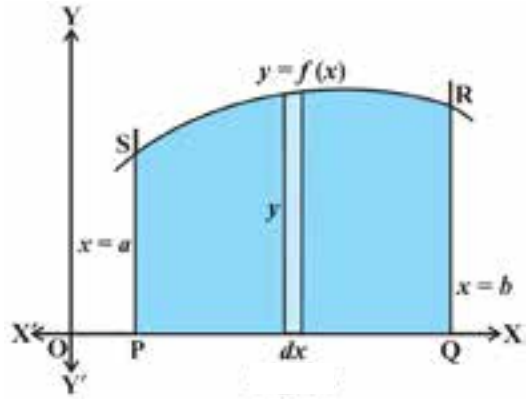
(a) 72

(b) 80

(c) 112

(d) 128

17. In the given figure, the shaded region represents which part of the circle?



(a) A quarter of the circle in the first quadrant bounded by OA, OB, and the arc AB

(b) A semicircle bounded by the X-axis and Y-axis

(c) The entire circle centered at O with radius a

(d) A quarter of the circle in the second quadrant bounded by OA, OB, and the arc AB

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18. A mapping $f: n \to N$, where N is the set of natural numbers is define as

$$f(n) = \begin{cases} n^2, & \text{for } n \text{ odd} \\ 2n+1, & \text{for } n \text{ even} \end{cases}$$

for $n \in N$. Then, f is

(a) Subjective but not injective

(b) Injective but not subjective

(c) Bijective

(d) neither injective nor subjective

19. **Assertion:** $\int \sin 3x \cos 5x \, dx = \frac{-\cos 8x}{16} + \frac{\cos 2x}{4} + C$

Reason: $2\cos A \sin B = \sin(A+B) - \sin(A-B)$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

20. **Assertion:** If $x = at^2$ and $y = 2at$, then $\left.\frac{d^2y}{dx^2}\right|_{t=2} = \frac{-1}{16a}$

Reason: $\frac{d^2y}{dx^2} = \left(\frac{dy}{dt}\right)^2 \times \left(\frac{dt}{dx}\right)^2$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Find a vector in the direction of vector $2\hat{i} - 3\hat{j} + 6\hat{k}$ which has magnitude of 21 units.

OR

Find the angle between X-axis and the vector $\hat{i} + \hat{j} + \hat{k}$.

22. Evaluate $\int \cos^{-1}(\sin x)\, dx$.

OR

Write the anti-derivative of $\left(3\sqrt{x} + \frac{1}{\sqrt{x}}\right)$.

23. What are the direction consines of a line which makes equal angles with the coordinate axes?

24. If $A = \{1,2,3\}, B = \{4,5,6,7\}$ and $f = \{(1,4),(2,5),(3,6)\}$ is a function from A to B. State whether f is one-one or not.

25. Two groups are computing for the positions of the Board of Directors of a corporation. The probabilities that the first and second groups will win are 0.6 and 0.4 respectively. Further, if the first group wins, the probability of introducing a new product introduced way by the second group.

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. The volume of a sphere is increasing at the rate of 3 cubic centimeter per second. Find the rate of increase of its surface area, when the radius is 2 cm.

OR

Show that the function $f(x) = 4x^3 - 18x^2 + 27x - 7$ is always increasing on R.

27. Evaluate $\int \frac{2\cos x}{\sin^2 x}\, dx$.

28. Find the area of a parallelogram whose adjacent sides represented by the vectors $2\hat{i} - 3\hat{k}$ and $4\hat{i} + 2\hat{k}$.

29. Using the principal values, write the value of $\cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right)$.

30. If $y = \sin^{-1}(6x\sqrt{1-9x^2})$, $-\frac{1}{3\sqrt{2}} < x < \frac{1}{3\sqrt{2}}$, then find $\frac{dy}{dx}$.

OR

If $(\cos x)^y = (\cos y)^x$, then find $\frac{dy}{dx}$.

31. If $\begin{vmatrix} x & \sin\theta & \cos\theta \\ -\sin\theta & -x & 1 \\ \cos\theta & 1 & x \end{vmatrix} = 8$, write the value of x.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

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32. Show that the lines $\frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7}$ and $\frac{x-2}{1}=\frac{y-4}{3}=\frac{z-6}{5}$ intersect. Also, find their point intersection.

33. Find $\int \frac{x^2+x+1}{(x+2)(x^2+1)}dx$.

OR

Evaluate $\int_0^{\pi} e^{2x}\cdot \sin\left(\frac{\pi}{4}+x\right)dx$.

34. Solve the differential equation $\frac{dy}{dx}-3y\cot x=\sin 2x$, given $y=2$ when $x=\frac{\pi}{2}$.

35. Maximize $Z=3x+2y$ subject to $x+2y\leq 0$, $3x+y\leq 15$, x, $y\geq 0$.

OR

Minimise $Z=x+2y$ subject to $2x+y\geq 3$, $x+2y\geq 6$, x, $y\geq 0$. Show that the minimum of Z occurs at more than two points.

SECTION - E

Case study based questions are compulsory.

36. Ravindra Manch was established in 1963 to commemorate the 100th birth anniversary of Ravindra Nath Tagore. Ravindra Manch is one of the myriad places in Jaipur that hold a historical significance. The auditorium was among the seventeen cultural centers that were envisioned by Pandit Jawaharlal Nehru and was thrown open to the public on Independence Day in the year 1963. Since then, the place has hosted a wide number of cultural shows and events. Some of the most renowned artists, dancers and actors have displayed their talent at this prestigious venue.



Last year, 300 people attended the Ravindra Manch Drama Club's winter play. The ticket price was ₹ 70. The advisor estimates that 20 fewer people would attend for each ₹ 10 increase in ticket price.

(i) What ticket price would give the most income for the Drama Club?

(ii) If the Drama Club raised its tickets to this price, how much income should it expect to bring in?

37. Cross holding, also referred to as cross shareholding, describes a situation where one publicly-traded company holds a significant number of shares of another publicly-traded company. The shares owned of the second publicly-traded company are referred to as a cross-holding of the first company.



Two companies A and B are holding shares in each other. A is holding 20% shares of B and B is holding 10% shares. of A. The separately earned profits of the two companies are ₹ 98000 and ₹ 49000 respectively.

(i) Find total profit of each company using matrix notations.

(ii) Show that the total of the profits allocated to outside shareholders is equal to the total of separately earned profit.

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38. Federal health officials have reported that the proportion of children (ages 19 to 35 months) who received a full series of inoculations against vaccine-preventable diseases, including diphtheria, tetanus, measles, and mumps, increased up until 2006, but has stalled since. The CDC reports that 14 states have achieved a vaccination coverage rate of at least 80% for the 4:3:1:3:3:1 series.26 The probability that a randomly selected toddler in Alabama has received a full set of inoculations is 0.792, for a toddler in Georgia, 0.839, and for a toddler in Utah, 0.711.27 Suppose a toddler from each state is randomly selected.



(i) Find the probability that all three toddlers have received these inoculations.

(ii) Find the probability that none of the three has received these inoculations.

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